

Beyond PII Removal: Risk-Budgeted Generalization for LLM-Ready Sensitive Text

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Abstract

1 Sensitive text corpora are valuable for training, retrieval, evaluation, and
 2 post-training adaptation of foundation-model systems, but common trans-
 3 formation pipelines still focus on direct personally identifying information
 4 (PII) such as names, emails, phone numbers, and addresses. This span-
 5 level view is necessary but incomplete: a record with no explicit name
 6 may still reveal sensitive attributes through quasi-identifiers such as exact
 7 dates, fine-grained locations, rare occupations, schools, medical or legal
 8 facts, family relations, citizenship cues, or combinations of ordinary details.
 9 We introduce Risk-Budgeted Quasi-Identifier Generalization (RB-QIG), a
 10 lightweight and auditable transformation layer that scores quasi-identifiers
 11 by privacy risk and utility importance, applies the least destructive step
 12 in a pre-specified generalization ladder, and stops when a document-level
 13 risk budget is met. On a controlled synthetic benchmark, RB-QIG balanced
 14 reduces risk-weighted leakage to 23.6% while preserving 71.7% of utility
 15 facts; blanket quasi-identifier redaction removes leakage but preserves only
 16 43.3% of utility facts. On a 100-record English RAT-Bench pilot, direct redac-
 17 tion leaves 78.0% LLM-attacker risk-weighted leakage and a naive LLM
 18 sanitizer leaves 46.0%, while RB-QIG balanced reduces leakage to 5.7%.
 19 We conclude that responsible data transformation for foundation-model
 20 pipelines should measure what capable models can still infer, not only
 21 which explicit PII spans were removed.

22 1 Introduction

23 Sensitive text is central to many data sources that would make foundation-model systems
 24 more useful: clinical notes, legal decisions, financial records, support logs, internal work-
 25 place messages, student advising records, and user-assistant conversations. Such data can
 26 support retrieval-augmented generation, model evaluation, safety auditing, preference-data
 27 construction, and domain adaptation. The same data also contains details that people did
 28 not intend to make public. Responsible data enablement therefore often begins with de-
 29 identification: remove names, replace emails, mask phone numbers, and redact addresses.

30 This direct-identifier pipeline is important, but it is not sufficient for LLM-ready text. A
 31 sentence such as “a 34-year-old oncology infusion pharmacist at a small startup in Asheville
 32 called about a brother’s rare condition after an April 2025 surgery” may contain no name
 33 or email after redaction, yet still expose a highly specific profile. Traditional privacy work
 34 formalized this problem through linkage and quasi-identifiers [Sweeney \(2002\)](#); [Machanava-](#)
 35 [jjhala et al. \(2007\)](#); [Li et al. \(2007\)](#); [Narayanan & Shmatikov \(2008\)](#). Modern language models
 36 make the problem more acute because an attacker need not match exact database columns:
 37 LLMs can combine paraphrase recognition, world knowledge, and contextual inference
 38 to recover sensitive attributes [Staab et al. \(2024\)](#). Recent text-anonymization benchmarks
 39 similarly argue that anonymization should be evaluated by residual re-identification risk,
 40 not only by span recall [Krčo et al. \(2026\)](#).

41 The practical question is therefore not just “did the tool remove obvious PII?” but “what can
 42 a capable model still infer, and how much useful meaning was preserved?” A conservative
 43 answer is to redact every quasi-identifier: occupations become [OCCUPATION], locations

become [LOCATION], dates become [DATE], and diagnoses become [MEDICAL]. This can reduce leakage, but it also destroys much of the information that made the corpus useful. The opposite extreme, direct PII removal only, preserves utility but leaves large residual inference risk.

We study the space between these extremes through Risk-Budgeted Quasi-Identifier Generalization (RB-QIG). RB-QIG does not train a model and does not rewrite whole documents. It takes candidate quasi-identifier spans, assigns each a privacy-risk score and a utility-importance score, and applies span-level replacements from a small ladder: keep, generalize once, generalize further, or redact. The algorithm greedily chooses the next edit that gives the largest risk reduction per unit of utility loss, stopping when a document-level risk budget is satisfied. The resulting audit log records which fact was generalized, which replacement was used, and why.

Our contributions are:

- a quasi-identifier taxonomy for LLM-ready sensitive text;
- a risk-budgeted generalization policy that operationalizes an inspectable privacy-utility tradeoff without training or fine-tuning;
- an evaluation protocol combining deterministic leakage metrics, LLM attackers, and LLM utility judges; and
- an analysis showing that residual leakage often comes from relational or lexical implications rather than missed exact spans.

We show that direct PII redaction and naive LLM sanitization leave substantial residual quasi-identifier risk, while RB-QIG sharply reduces that measured risk and exposes a useful controlled privacy-utility frontier.

2 Background and Threat Model

From tabular quasi-identifiers to text. Classical privacy work showed that attributes such as ZIP code, gender, and birth date can link a record to a person even when names are absent [Sweeney \(2002\)](#). l -diversity and t -closeness refined this view by showing that indistinguishability alone may not protect sensitive attributes [Machanavajjhala et al. \(2007\)](#); [Li et al. \(2007\)](#). Differential privacy offers a different guarantee for statistical releases [Dwork et al. \(2006\)](#), but it does not directly solve the instance-level text-transformation problem studied here: preserve a useful text record while reducing inferable identifiers inside that record.

Text de-identification. Applied de-identification systems often target named entities and regulated identifier categories. Clinical de-identification has a long history of shared tasks and neural sequence-labeling systems [Uzuner et al. \(2007\)](#); [Dernoncourt et al. \(2017\)](#). Practical tools such as Presidio support PII detection, masking, and replacement [Microsoft \(2026\)](#); specialized privacy filters provide another local PII-masking building block [OpenAI \(2026\)](#). These systems are useful, but they are typically optimized for identifiable spans rather than the residual inference risk created by combinations of demographic, relational, narrative, and domain facts. TAB provides a legal-domain anonymization benchmark [Pilán et al. \(2022\)](#), and RAT-Bench directly emphasizes LLM-based re-identification risk from direct and indirect identifiers [Krčo et al. \(2026\)](#).

LLM privacy leakage. LLMs introduce two complementary privacy concerns. Training data may be memorized or extractable under some conditions [Carlini et al. \(2021; 2023\)](#). Separately, even when a private fact is not explicit, a model may infer it from surrounding text. Staab et al. show that LLMs can infer personal attributes from ordinary text in their setting [Staab et al. \(2024\)](#). RB-QIG focuses on the second concern at the transformation stage: before using sensitive text for retrieval, evaluation, or post-training, reduce what a capable model can infer from the transformed record.

Table 1: Quasi-identifier categories used by RB-QIG. The examples are illustrative; the implementation stores type, risk, utility, and a replacement ladder for each candidate span.

Category	Example precise value	Safer generalized value
Date/age	April 17, 2025; 34	month/year; 30s
Location	Asheville, NC	city in the southeastern U.S.
Occupation	oncology infusion pharmacist	clinical pharmacy professional
Institution	named school or employer	university; healthcare company
Medical/legal fact	rare condition or case event	broader condition/event type
Family relation	younger brother under guardianship	dependent family member
Demographic cue	citizenship or heritage phrase	broader demographic placeholder
Narrative cue	unique life event	coarse role-preserving description

Threat model. The attacker receives transformed text and, in the LLM-attacker setting, a list of benchmark attributes to infer. The attacker returns structured guesses with short evidence. We report exact recovery, coarse recovery, record compromise, and risk-weighted leakage. We use only synthetic profiles and public benchmarks; we do not attempt to identify real people. This threat model does not cover arbitrary auxiliary databases or determined human investigators. It is a reproducible stress test for whether a transformation leaves sensitive attributes inferable.

We distinguish three evaluation settings. In the *oracle/controlled* setting, ground-truth quasi-identifier spans are known from synthetic data. In the *target-aware public* setting, RAT-Bench target attributes are used to find value-revealing spans; this isolates transformation quality when the sensitive attributes of interest are known. In the *blind public* setting, the extractor does not see target attributes at extraction time and is scored against benchmark annotations afterward. The blind setting is closest to deployment.

3 Risk-Budgeted Quasi-Identifier Generalization

3.1 Taxonomy and Generalization Ladders

RB-QIG uses typed candidate spans. The taxonomy is intentionally broad because LLM inference often uses facts that are not direct PII. Table 1 summarizes the categories used in the prototype.

A generalization ladder is a short ordered list of replacements. For example, “Boise, Idaho” may map to “a city in the U.S. Mountain West,” then “a city in the western United States,” and finally [LOCATION]. The ladder is not learned; it is a transparent policy artifact. This matters for sensitive data owners because they may want to inspect, revise, or prohibit particular replacement types before releasing transformed text.

3.2 Risk and Utility Scores

Each quasi-identifier q_i receives a privacy-risk score $r_i \in [0, 5]$ and a utility-importance score $u_i \in [0, 5]$. Risk reflects specificity, rarity, sensitivity, and linkability. Utility reflects whether the fact is needed for the benign task: diagnosis category may be important in a clinical triage note, while an exact date of birth is usually not. The current prototype uses transparent heuristics and benchmark metadata rather than learned risk models. This is a limitation, but it also makes the method auditable.

The document risk score is

$$R(x) = \sum_i r_i + 0.5 \binom{|\{i : r_i \geq 3\}|}{2},$$

where the second term approximates combination risk from multiple linkable high-risk attributes. Candidate edits are evaluated by their risk drop divided by utility loss. RB-QIG greedily applies the edit with the best ratio until $R(x)$ falls below the budget or no edits

127 remain. All edits are span replacements, so the system avoids unconstrained paraphrase
128 hallucination.

129 3.3 Baselines

130 We compare against five baselines. *None* leaves the record unchanged. *Direct redaction*
131 removes direct identifiers using regexes and benchmark-provided direct spans. *Naive LLM*
132 *direct* asks an LLM sanitizer to remove personally identifying information while preserving
133 meaning. *Blanket QI redaction* replaces every quasi-identifier with a typed placeholder.
134 *Presidio-style pattern redaction* is a practical pattern baseline included as a lower-bound
135 diagnostic rather than a full production Presidio deployment. RB-QIG is evaluated with
136 strict, balanced, and utility-oriented budgets in the synthetic setting, with the balanced
137 policy as the main public method.

138 4 Experiments

139 **Synthetic control.** We generate 100 synthetic records across medical, legal, financial,
140 workplace, and education-support domains. Each record has direct identifiers, quasi-
141 identifiers, ground-truth attributes, a task label, and utility facts. Because oracle quasi-
142 identifier spans are known, this setting isolates the transformation tradeoff: given the
143 sensitive spans, does risk-budgeted generalization preserve more useful information than
144 blanket redaction?

145 **RAT-Bench pilot.** We evaluate 100 English difficulty-1 RAT-Bench records Krčo et al.
146 (2026). The target-aware version uses benchmark target attributes to identify value-revealing
147 spans and variants. We also run a blind extractor over the same 100 records and score it
148 against benchmark quasi-identifiers. A deterministic backstop for common demographic,
149 citizenship, education, employment, marital, and race/ethnicity cues improves coverage.

150 **TAB legal screen.** We run a 30-document TAB ECHR legal-domain screen Pilán et al.
151 (2022). The TAB experiment is intentionally bounded and deterministic. It tests whether the
152 residual-risk story transfers to another domain.

153 **LLM attacker and utility judge.** For RAT-Bench, an LLM attacker receives transformed
154 text and requested attribute names, then returns structured guesses with evidence. An
155 LLM utility judge scores label preservation, fact preservation, and semantic utility without
156 rewarding preserved direct identifiers. We also run bounded stronger-attacker checks with
157 GPT-5.5 and a mini-model agreement check on the blind public outputs. All API calls use
158 synthetic or public benchmark records.

159 **Metrics.** Privacy metrics are direct-identifier leakage, exact attribute leakage, coarse at-
160 tribute leakage, record compromise, and risk-weighted leakage. Utility metrics are utility-
161 fact preservation, label preservation, semantic utility, annotation-derived quasi-identifier
162 specificity, and token change.

163 5 Results and Analysis

164 **Controlled privacy–utility frontier.** Table 2 shows the cleanest evidence for the method.
165 Direct redaction preserves utility but leaves all quasi-identifiers. Blanket quasi-identifier
166 redaction eliminates measured leakage but preserves only 43.3% of utility facts. RB-QIG
167 balanced preserves 71.7% of utility facts while reducing risk-weighted leakage from 100.0%
168 to 23.6%. This supports the controlled claim that risk-budgeted generalization can occupy a
169 useful middle point between direct redaction and blanket redaction.

170 **Semantic utility is modest.** The LLM utility judge in Table 4 is more forgiving than
171 the synthetic fact metric because broad placeholders can still preserve some document

Table 2: Synthetic controlled benchmark.

Method	Risk-weighted leak	Utility facts	Token change
None	100.0	100.0	0.0
Direct	100.0	100.0	13.5
Blanket QI	0.0	43.3	63.7
RB-QIG strict	16.1	48.3	59.3
RB-QIG balanced	23.6	71.7	59.0
RB-QIG utility	38.0	95.0	56.6

Table 3: RAT-Bench LLM attacker results.

Method	Record compromise	Exact attr leak	Coarse attr leak	Risk-weighted leak
Direct	76.0	74.9	81.2	78.0
LLM direct	34.0	48.7	52.8	46.0
Blanket QI	3.0	3.9	7.3	5.3
RB-QIG balanced	2.0	4.0	8.8	5.7

172 meaning. Under this metric, RB-QIG balanced has only a modest semantic utility edge
 173 over blanket redaction: +2.6 points. Local placeholder, fluent-rewrite, and attribute-level
 174 fluent diagnostics remove surface artifacts or improve specificity, but do not rescue this
 175 public-utility gap. A text-quality audit further suggests that phrase-level surface cleanup is
 176 not the missing ingredient: the lowest-artifact attr-fluent variant has the worst hidden-label
 177 utility, and longer or more generalized replacements correlate negatively with utility in the
 178 pooled audit. A same-text audit also finds that the method-labeled privacy-aware judge is
 179 noisy for fine-grained rankings.

180 **Direct redaction leaves high residual LLM risk.** Table 3 shows that direct redaction is
 181 insufficient on the RAT-Bench pilot. After direct redaction, the LLM attacker still recovers
 182 many target attributes: 78.0% risk-weighted leakage. The naive LLM sanitizer improves
 183 over direct redaction but still leaves 46.0% leakage. RB-QIG balanced reduces leakage to
 184 5.7%, a paired drop of 72.2 points relative to direct redaction and 40.2 points relative to the
 185 LLM sanitizer. Thus the privacy claim is reduction relative to direct and naive redaction,
 186 not superiority over blanket redaction.

187 **Blind extraction is the deployment bottleneck.** Table 5 summarizes the blind public stress
 188 test. The raw blind extractor covers 72.8% of benchmark QI spans; a generic deterministic
 189 backstop raises coverage to 99.6%. Under the LLM attacker, blind-backstop RB-QIG
 190 balanced reduces risk-weighted leakage from 78.1% for direct redaction to 6.4%, and is tied
 191 with blanket redaction. Deterministic blind scoring gives the same direct-redaction failure
 192 but is harsher on generalized cues: direct redaction leaves 94.3%, RB-QIG balanced leaves
 193 5.4%, the direct-to-RB-QIG drop is 88.9 points.

194 **Stronger attackers and budget diagnostics.** A 50-record GPT-5.5 attacker check is harsher
 195 than the low-cost attacker: direct redaction leaves 87.8% risk-weighted leakage, RB-QIG
 196 balanced leaves 31.9%, and blanket QI leaves 29.7%. The direct-to-RB-QIG reduction remains
 197 large. A 20-record blind mini-model check gives the same qualitative ordering: direct 75.0%,
 198 RB-QIG balanced 14.8%, and blanket QI 16.8%. Pairwise exact/coarse leak agreement is
 199 moderate: 83.3%/80.0% for mini versus GPT-5.5, and 76.7%/71.9% for nano versus GPT-5.5,
 200 so attacker strength affects absolute leakage but not the qualitative ordering. budget-frontier
 201 diagnostics show the control knob is meaningful: on blind RAT-Bench, budgets 2/4/6
 202 give 4.4%/5.4%/7.2% deterministic risk-weighted leakage and 30.1%/33.2%/36.4% quasi-
 203 identifier specificity; on TAB they give 7.8%/12.3%/20.4% leakage and 28.1%/31.2%/34.3%
 204 specificity. Annotation-derived specificity is higher for RB-QIG balanced than blanket by
 205 +6.8 points on blind RAT-Bench and +6.2 on TAB. Additional target-aware smokes preserve
 206 the same ordering on difficulty-2 (direct 70.8% and naive sanitizer 56.9% versus RB-QIG

Table 4: Synthetic LLM utility judge results.

Method	Label preserved	LLM fact preservation	Semantic utility
Direct	100.0	75.3	83.2
Blanket QI	99.0	69.1	76.8
RB-QIG balanced	100.0	71.3	79.4

Table 5: Blind public RAT-Bench stress test.

Method	LLM risk-weighted leak	Semantic utility	LLM fact preservation
Direct	78.1	86.2	84.3
Blanket QI	6.8	62.6	53.8
RB-QIG balanced	6.4	62.2	55.5

13.1%) and on the 20-record target-aware mini smoke (RB-QIG 18.0% versus 77.3% for direct).

Cross-domain and practical-baseline diagnostics. The TAB screen provides deterministic cross-domain support: direct redaction leaves 99.8% risk-weighted leakage, while RB-QIG balanced lowers it to 12.3%. Its bounded 20-document legal-task utility screen shows that blanket scores 73.0%, while RB-QIG balanced scores 71.0%. The Presidio-style pattern-only baseline gives a practical lower-bound diagnostic: on RAT-Bench it leaves 37.0% direct-identifier leakage and 85.0% deterministic risk-weighted leakage; on TAB it misses legal names and case identifiers, leaving 100.0% direct-identifier leakage. This makes the oracle-assisted direct baseline conservative rather than weak.

Failure taxonomy. Residual RB-QIG leaks are mostly relational or lexical variants: widowhood from bereavement cues, sex from gendered language, education from school-name mentions, citizenship from non-citizen phrasing, and employment or armed-forces status from occupation context. These failures are not just missed exact spans; they are cues that a model can combine. This supports the broader argument that data transformation should evaluate inferability, not only span categories.

6 Limitations and Ethics

RB-QIG is not an anonymization guarantee and should not be used as a legal or compliance claim. The main public benchmark extraction is target-aware, so it evaluates transformation behavior under known benchmark attributes rather than fully blind deployment. All experiments use synthetic or public benchmark data. We do not attempt to identify real people or infer private attributes about individuals.

7 Conclusion

Direct PII removal is an incomplete transformation for LLM-ready sensitive text. RB-QIG demonstrates that a small, auditable risk-budgeted generalization layer can dramatically reduce measured residual inference risk while preserving useful semantic content better than blanket redaction in controlled settings. The main open problem is robust blind quasi-identifier extraction and rewriting that preserves task-relevant utility while handling relational, lexical, and world-knowledge inference cues. More broadly, responsible data enablement should evaluate what capable models can still infer, not only what explicit PII spans were removed.

Table 6: Representative residual failure categories.

Failure type	Example cue	Why span removal misses it	Possible fix
Relation cue	family status or be-reavement phrase	sensitive status is inferred from a relation, not named directly	relation-aware rewrite
Gender cue	pronouns or gendered family roles	sex can be inferred after direct identifiers are removed	neutralization pass
Cultural cue	heritage, tradition, or language mention	race or ethnicity may be inferred from ordinary context	coarser cultural phrase
Institution cue	school, employer, or service branch	education or employment is exposed by a named institution	institution ladder
Placeholder cue	typed demographic placeholder in context	the placeholder itself can reveal the attribute class	neutral placeholders

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